

BEGINNER'S HANDBOOK

Git, GitHub, SSH & VS Code

A complete, zero-to-hero setup guide. Install everything, connect securely with SSH, and make your first commit — explained step by step for absolute beginners.

Windows

macOS

Linux

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1. Understanding the Big Picture

Before touching a keyboard, it helps to know what each tool actually does and how they fit together.

Tool	What it is	Why you need it
Git	A version control program that runs on your computer	Tracks every change you make to your code, so you can undo mistakes and work with others
GitHub	A website that hosts Git projects online	Lets you back up your code in the cloud, share it, and collaborate
SSH	A secure way for your computer to "log in" to GitHub	Lets you push/pull code without typing a password every time
VS Code	A code editor	Where you write code, and where Git actions are built in with a friendly interface

Think of it like this:

Git is the engine that records history. GitHub is the garage in the cloud where you park your project. SSH is the key that lets you into that garage without shouting your password every time. VS Code is the workshop where you actually build things.

The order we'll set things up in: **Git → GitHub account → SSH key → VS Code → first project**. Each step only takes a few minutes.

2. Installing Git

WINDOWS

- 1 Go to `git-scm.com/download/win` in your browser. The download starts automatically.
- 2 Run the installer. Keep clicking **Next** and accept the default options — they're fine for beginners. On the "Adjusting your PATH" screen, keep the recommended option selected.
- 3 Once installed, open **Git Bash** from the Start Menu (it was installed alongside Git). This gives you a terminal that understands Git and Linux-style commands.
- 4 Verify the install by typing:

```
git --version
```

You should see something like `git version 2.45.0`.

MACOS

- 1 Open the **Terminal** app (press `Cmd + Space`, type "Terminal", hit Enter).
- 2 Type the command below. macOS will offer to install Git for you as part of the Xcode Command Line Tools.

```
git --version
```

If Git isn't installed yet, a popup appears — click **Install** and wait a few minutes.

- 3 (Optional, recommended) Install [Homebrew](#) first, then install the latest Git version through it:

```
/bin/bash -c "$(curl -fsSL https://raw.githubusercontent.com/Homebrew/install/HEAD/install.sh)"  
brew install git
```

LINUX

- 1 Open your terminal and run the command for your distribution:

```
# Debian / Ubuntu  
sudo apt update && sudo apt install git -y  
  
# Fedora  
sudo dnf install git -y  
  
# Arch  
sudo pacman -S git
```

Tell Git who you are

Every commit you make is stamped with a name and email. Set this once, on any operating system, using your terminal / Git Bash:

```
git config --global user.name "Your Name"  
git config --global user.email "your.email@example.com"
```

Tip

Use the same email address you'll use for your GitHub account — it makes your commits show up correctly linked to your profile later.

3. Creating Your GitHub Account

- 1 Go to `github.com` and click **Sign up**.
- 2 Enter an email, create a password, and choose a username. Pick something professional — this can end up on your resume.
- 3 Verify your email using the code GitHub sends you.
- 4 Choose the **Free** plan when asked — it covers everything a beginner (and most professionals) need.
- 5 *(Recommended)* Turn on two-factor authentication: Settings → Password and authentication → Two-factor authentication.

Important

GitHub no longer accepts your account password for Git operations over HTTPS. This is exactly why we set up SSH in the next section — it's the modern, secure way to connect.

4. Setting Up SSH for GitHub

SSH keys come in a pair: a **private key** (stays secret on your computer, never shared) and a **public key** (uploaded to GitHub). GitHub uses this pair to confirm it's really you, without a password.

Step 1 — Check for existing keys

Open Terminal (macOS/Linux) or Git Bash (Windows) and run:

```
ls -al ~/.ssh
```

If you see files named `id_ed25519` and `id_ed25519.pub` (or `id_rsa` / `id_rsa.pub`), you already have a key and can skip to Step 3.

Step 2 — Generate a new SSH key

```
ssh-keygen -t ed25519 -C "your.email@example.com"
```

- When asked "Enter a file in which to save the key," just press **Enter** to accept the default location.
- When asked for a passphrase, you can press **Enter** twice for no passphrase, or set one for extra security (recommended). If you set one, you'll be asked for it occasionally.

Step 3 — Start the SSH agent and add your key

```
# macOS / Linux
eval "$(ssh-agent -s)"
ssh-add ~/.ssh/id_ed25519

# Windows (Git Bash)
eval $(ssh-agent -s)
ssh-add ~/.ssh/id_ed25519
```

Step 4 — Copy your public key

```
# macOS
pbcopy < ~/.ssh/id_ed25519.pub

# Windows (Git Bash)
cat ~/.ssh/id_ed25519.pub | clip

# Linux (needs xclip installed)
xclip -sel clip < ~/.ssh/id_ed25519.pub
```

No clipboard tool? Just open the file and copy the text manually:

```
cat ~/.ssh/id_ed25519.pub
```

Step 5 — Add the key to GitHub

- 1 On GitHub, click your profile photo → **Settings**.
- 2 In the left sidebar, click **SSH and GPG keys**.
- 3 Click **New SSH key**. Give it a title (e.g. "My Laptop"), paste the key into the "Key" field, and click **Add SSH key**.

Step 6 — Test the connection

```
ssh -T git@github.com
```

The first time, you'll see a warning about an unknown host — type `yes` and press Enter. A successful connection replies with:

```
Hi your-username! You've successfully authenticated, but GitHub does not
provide shell access.
```

That message is a good sign

Seeing "does not provide shell access" is expected and means SSH is working correctly — GitHub isn't a computer you log into, just a service you authenticate with.

5. Installing & Configuring VS Code

- 1
- Go to `code.visualstudio.com` and click the big **Download** button — it detects your operating system automatically.
- 2
- Run the installer.
 - **Windows:** during setup, check the boxes for "Add to PATH" and "Add 'Open with Code' action."
 - **macOS:** drag VS Code into the Applications folder, then open it once from there.
 - **Linux:** install the `.deb` / `.rpm` package, or use your package manager (e.g. `sudo snap install code --classic`).
- 3
- Open VS Code and install the essentials from the Extensions panel (icon of four squares on the left sidebar, or `Ctrl/Cmd + Shift + X`):

Extension	Why install it
GitLens	See who changed each line, browse history, compare branches visually
GitHub Pull Requests	Review and manage GitHub pull requests without leaving the editor
Prettier / language-specific packs	Auto-formatting for the language you're working in

Connect VS Code to your GitHub account

- 1
- Click the Accounts icon (bottom-left corner, person silhouette).
- 2
- Choose **Sign in with GitHub** and follow the browser prompt to authorize VS Code.

This isn't required for SSH-based Git operations to work, but it enables handy features like GitHub Pull Requests and Codespaces later.

Built-in terminal

VS Code has its own terminal panel: `View → Terminal` or `Ctrl/Cmd + ``. All the Git commands in this guide work exactly the same there.

6. Your First Project: Clone, Commit, Push

Create a repository on GitHub

- 1 On GitHub, click the + icon (top-right) → **New repository**.
- 2 Name it (e.g. `my-first-repo`), leave it Public or set to Private, and check "**Add a README file.**" Click **Create repository**.

Clone it to your computer using SSH

- 1 On the repository page, click the green **Code** button, select the **SSH** tab, and copy the address (looks like `git@github.com:username/my-first-repo.git`).
- 2 In VS Code, press `Ctrl/Cmd + Shift + P`, type **Git: Clone**, paste the URL, and choose a folder to save it in. Open the folder when prompted.

Or from any terminal:

```
git clone git@github.com:username/my-first-repo.git
cd my-first-repo
```

Make a change and commit it

- 1 Edit the README, or create a new file, in VS Code.
- 2 Open the Source Control panel (icon with the branching lines, or `Ctrl/Cmd + Shift + G`). You'll see your changed file listed.
- 3 Hover the file and click the + to stage it, type a short message describing the change in the message box, then click the checkmark **Commit** button.
- 4 Click **Sync Changes** (or **Push**) to send your commit up to GitHub.

The equivalent terminal commands, if you'd rather type them:

```
git add .
git commit -m "Describe what you changed"
git push
```

Refresh the repository page on GitHub — your change is now live in the cloud.

7. Everyday Git Cheat Sheet

Command	What it does
<code>git status</code>	Shows what's changed and what's staged
<code>git add <file></code>	Stages a specific file for commit
<code>git add .</code>	Stages every changed file
<code>git commit -m "message"</code>	Saves a snapshot with a description
<code>git push</code>	Uploads commits to GitHub
<code>git pull</code>	Downloads and merges the latest changes
<code>git clone <url></code>	Downloads a repository for the first time
<code>git branch new-feature</code>	Creates a new branch
<code>git checkout new-feature</code>	Switches to that branch
<code>git log --oneline</code>	Shows a compact commit history

Your first-time setup checklist

- ☐ Git installed and `git --version` works
- ☐ `user.name` and `user.email` configured
- ☐ GitHub account created and verified
- ☐ SSH key generated and added to GitHub
- ☐ `ssh -T git@github.com` greets you by username
- ☐ VS Code installed with GitLens
- ☐ First repository cloned, edited, committed, and pushed

8. Troubleshooting

"Permission denied (publickey)" when pushing or pulling

Your SSH key isn't loaded or wasn't added to GitHub. Re-run `ssh-add ~/.ssh/id_ed25519` and confirm the public key is listed under GitHub → Settings → SSH and GPG keys.

"git: command not found"

Git isn't installed correctly, or your terminal wasn't restarted after installing. Close and reopen the terminal, then try `git --version` again.

Git asks for a username and password instead of using SSH

You likely cloned using the HTTPS URL instead of SSH. Check your remote and switch it:

```
git remote -v
git remote set-url origin git@github.com:username/repo-name.git
```

VS Code doesn't show the Source Control changes

Make sure the folder you opened is the actual Git repository folder (it should contain a hidden `.git` folder), not a folder one level above or below it.

Wrong name/email showing on commits

Update your global config, then it applies going forward (past commits are unaffected):

```
git config --global user.name "Correct Name"
git config --global user.email "correct@email.com"
```

You now have everything set up: Git tracks your work, GitHub stores and shares it, SSH connects them securely, and VS Code ties it all together. Happy coding!